



15 MIN • TEAM PRESENTATION

ClaimGuard

Evidence-backed academic integrity checking

From PDF claims to reference validation and source-support verdicts

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Why academic integrity checking matters

A citation can be real...

**...and still not support
the sentence that cites
it.**

Manual review is slow; naive automation
creates false accusations.

TRIAGE, NOT A VERDICT

- **Scale** Long reports contain dozens of checkable claims.
- **Subtlety** Existence, relevance, and factual support are different questions.
- **Risk** False positives can unfairly damage students or researchers.
- **Need** Reviewers require traceable evidence, not a black-box score.

Goal: make suspicious passages faster to inspect and easier to explain.

Three checks — one auditable report

01

Does this need a citation?

Claim type, section, citation marker, severity

02

Does the source exist?

IEEE parsing plus scholarly metadata APIs

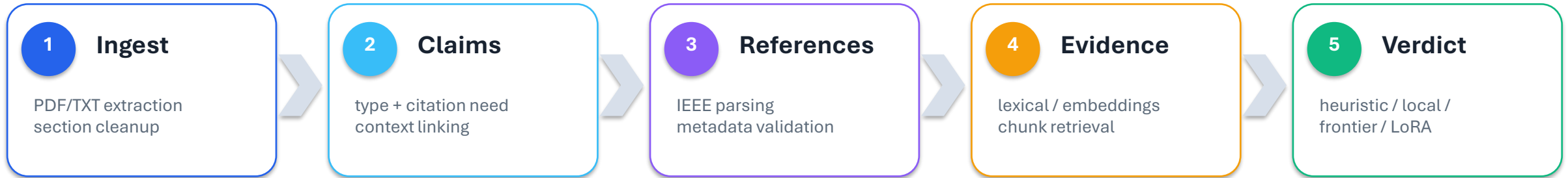
03

Does the source support it?

Evidence retrieval plus five-way verification

Output: human-readable Markdown + complete JSON provenance for evaluation and audit

Pipeline architecture

**AUDIT RAIL**

evidence • confidence • model ID • latency • token usage • response/request IDs

Offline first

Deterministic fallback

Network features are opt-in

Same schema

Comparable backends

Five support labels

Human review

Final decision

No automatic misconduct claim

Technical decisions and trade-offs

RETRIEVAL

all-MiniLM-L6-v2

Semantic matching; cosine fallback on Windows

METADATA

CrossRef • OpenAlex • S2

Coverage across DOI, title, authors, OA status

LOCAL MODEL

Ollama / qwen3:1.7b

Private and offline; slower on this machine

FRONTIER

OpenAI gpt-5.4-mini

Fast structured output; paid and external

FINE-TUNED

SciFact LoRA

Cheap local inference; three-label taxonomy

Design principle: graceful degradation — APIs or embeddings may fail without stopping the report.

Demo on our real RQ5 report

POWERSHELL

```
python run_claimguard.py `
  --input Reports/NW2_RQ5_Report.pdf --verifier openai `
  --output outputs/rq5_openai.json --markdown-output
  outputs/rq5_openai.md
```

137

sentences

65factual
claims**21**citation
flags

REAL CLAIM FROM SENTENCE 11

“RAG connects a language model to external knowledge without changing model weights.”

Citations detected: [1, 3] • Top evidence retained

OPENAI**insufficient
evidence**

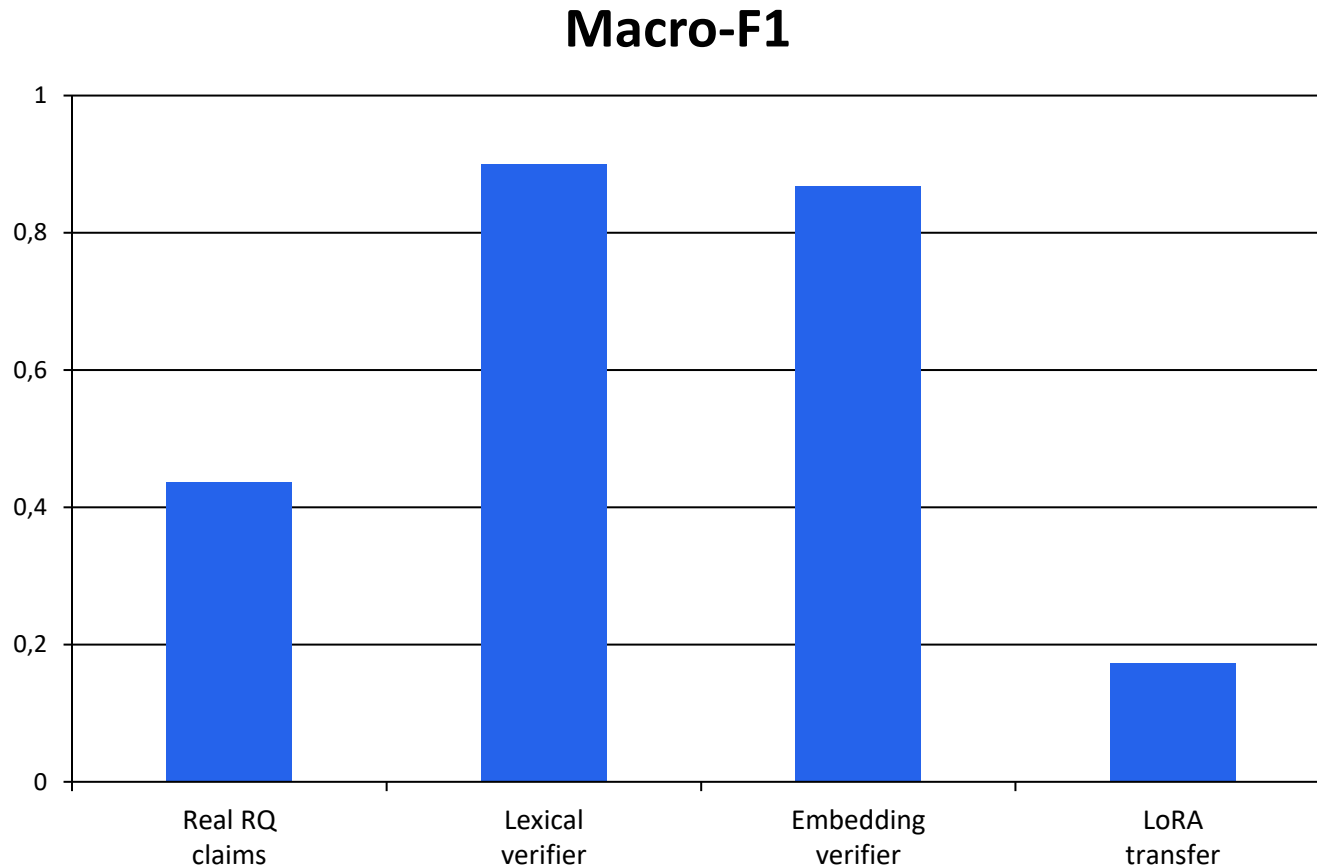
1.889 s • 506 tokens

OLLAMA**supported**

40.902 s • local

Demo message: the evidence is visible — disagreement becomes reviewable instead of hidden.

Key results: controlled tests vs. real reports

**1.000****IEEE parser accuracy**

5-field-complete test cases

0.267**Citation-needed F1**

50 manually labeled RQ sentences

0.353**SciFact validation Macro-F1**

450 validation pairs

Main finding: synthetic benchmarks overestimate how well simple rules transfer to authentic PDFs.

Model comparison: speed, control, and taxonomy

Backend	Observed result	Latency	Best use	Limitation
Heuristic	0.900 F1*	~0 ms	Transparent baseline	Relies on surface patterns
LoRA	0.172 F1*	9.3 ms	Cheap local inference	SciFact has 3 vs. 5 labels
Ollama	supported†	40.9 s	Private local review	Slow; one diagnostic claim
OpenAI	insufficient†	1.89 s	Structured frontier review	Paid/external; one diagnostic claim

* 30-case gold benchmark † same real RQ5 claim; diagnostic only — not an accuracy comparison

What works — and where it breaks

WORKS WELL

- End-to-end PDF → Markdown/JSON workflow
- Traceable evidence and backend metadata
- Robust IEEE parsing for Report 5
- Offline fallback and real CUDA LoRA path

LIMITATIONS

- PDF extraction introduces spacing/heading noise
- Real claim detection generalizes poorly
- API metadata may time out or rate-limit
- Binoculars' calibrated 14B pair exceeds 8 GB VRAM

Ethical boundary: a flag prioritizes human review; it never proves misconduct or AI authorship.

CONCLUSION

ClaimGuard makes integrity review auditable.

01

Pipeline

Claims, references, evidence, and verdicts in one workflow.

02

Evidence

Real-report evaluation exposes where synthetic scores mislead.

03

Responsibility

Use automation to focus human attention — not replace judgment.

Next: expand double-annotated real claims and run the full OpenAI/Ollama gold benchmark.



Questions & discussion

5 minutes

Architecture • Evaluation • Model trade-offs • Ethics